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# DBA5101 Analytics in Managerial Economics

## Homework 2

Ying WANG

### Problem 1

Given Information about amusement park customers:

- Thrill-Seekers (T):  $P_T = 70 - 2Q_T$
- Families (F):  $P_F = 50 - Q_F$
- Marginal Cost:  $MC = 10$

(a) The price under discrimination should be applied are showing below:

- Thrill-Seekers:  $P_T^* = 40$  SGD,  $Q_T^* = 15$
- Families:  $P_F^* = 30$  SGD,  $Q_F^* = 20$

The reasoning is as follows: for profit maximization, we apply the rule  $MR = MC$  for each market separately.

**Thrill-Seekers:**

$$\begin{aligned}P_T &= 70 - 2Q_T \\R_T &= P_T \cdot Q_T = (70 - 2Q_T)Q_T = 70Q_T - 2Q_T^2 \\MR_T &= \frac{d(R_T)}{dQ_T} = 70 - 4Q_T\end{aligned}$$

Here we set  $MR_T = MC$  to maximize revenue:

$$70 - 4Q_T = 10 \implies Q_T = 15$$

$$\text{Therefore: } P_T = 70 - 2 \times 15 = 70 - 30 = 40$$

**Families:**

$$\begin{aligned}P_F &= 50 - Q_F \\R_F &= P_F \cdot Q_F = (50 - Q_F)Q_F = 50Q_F - Q_F^2 \\MR_F &= \frac{d(R_F)}{dQ_F} = 50 - 2Q_F\end{aligned}$$

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Here we set  $MR_F = MC$  to maximize revenue:

$$50 - 2Q_F = 10 \implies Q_F^* = 20$$

$$\text{Therefore: } P_F^* = 50 - 20 = 30$$

(b) The uniform pricing should be applied is showing below:

- Uniform Price:  $P^* = 33.33$  SGD
- Total Quantity:  $Q_{total}^* = 35$  (Thrill-seekers: 18.33, Families: 16.67)

The reasoning is as follows: to find the aggregate demand, we need to sum quantities at each price level. Here set  $P_T = P_F = P$

$$P_T = 70 - 2Q_T \implies Q_T = \frac{70 - P}{2} \text{ (valid for } P \leq 70)$$

$$P_F = 50 - Q_F \implies Q_F = 50 - P \text{ (valid for } P \leq 50)$$

(i) When  $P \leq 50$ , both groups buy:

$$Q_{total} = Q_T + Q_F = \frac{70 - P}{2} + (50 - P) = 85 - 1.5P$$

Profit function:

$$\begin{aligned} \pi(P) &= (P - MC) \cdot Q_{total}(P) = (P - 10)(85 - 1.5P) \\ &= 85P - 1.5P^2 - 850 + 15P = 100P - 1.5P^2 - 850 \end{aligned}$$

Taking derivative with respect to P:

$$\frac{d\pi}{dP} = 100 - 3P$$

Setting  $\frac{d\pi}{dP} = 0$ :

$$100 - 3P = 0 \implies P^* = 33.33$$

Therefore:

$$\begin{aligned} Q_{total}^* &= 85 - 1.5 \times 33.33 = 85 - 50 = 35 \\ \pi &= (33.33 - 10) \times 35 = 23.33 \times 35 = 816.67 \end{aligned}$$

Individual quantities:

$$Q_T = \frac{70 - 33.33}{2} = 18.33$$

$$Q_F = 50 - 33.33 = 16.67$$

The price is valid since  $P^* = 33.33 \leq 50$ .

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(ii) When  $P > 50$ , only thrill-seekers buy:

$$Q_{total} = Q_T = \frac{70 - P}{2}$$

Profit function:

$$\begin{aligned}\pi(P) &= (P - MC) \cdot Q_T(P) = (P - 10) \cdot \frac{70 - P}{2} \\ &= \frac{(P - 10)(70 - P)}{2} = \frac{70P - P^2 - 700 + 10P}{2} \\ &= \frac{80P - P^2 - 700}{2}\end{aligned}$$

Taking derivative with respect to  $P$ :

$$\frac{d\pi}{dP} = \frac{80 - 2P}{2} = 40 - P$$

Setting  $\frac{d\pi}{dP} = 0$ :

$$40 - P = 0 \implies P^* = 40$$

Therefore:

$$\begin{aligned}Q_{total}^* &= \frac{70 - 40}{2} = 15 \\ \pi &= (40 - 10) \times 15 = 30 \times 15 = 450\end{aligned}$$

Since  $P^* = 40 \leq 50$ , this contradicts our assumption  $P > 50$ . Therefore, this case does not yield a valid solution in the range  $P > 50$ .

(iii) Boundary at  $P = 50$ :

At the transition point where families stop buying (the profit function has a kink due to piecewise demand):

$$\begin{aligned}Q_T &= \frac{70 - 50}{2} = 10, \quad Q_F = 50 - 50 = 0 \\ \pi &= (50 - 10) \times 10 = 400\end{aligned}$$

Comparing all three cases: Case (i) yields  $\pi = 816.67$ , Case (ii) is invalid, and Case (iii) yields  $\pi = 400$ . Therefore, the optimal uniform price is  $P^* = 33.33$  from Case (i).

(c) The total profits in the 2 scenarios are showing below:

- Profit with Price Discrimination: 850 SGD
- Profit with Uniform Pricing: 816.67 SGD
- Difference: 33.33 SGD (4.08% increase)

**Explanation:** Price discrimination is more profitable. Because it allows the park to charge higher prices to the less elastic group (thrill-seekers) while still capturing the more price-sensitive families market. The park extracts more consumer surplus by tailoring prices to each group's willingness to pay, resulting in approximately 4% higher profits compared to uniform pricing.

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The calculations are as follows:

**Profit with Price Discrimination:**

$$\begin{aligned}\pi_{PD} &= (P_T^* - MC) \cdot Q_T^* + (P_F^* - MC) \cdot Q_F^* \\ &= (40 - 10) \times 15 + (30 - 10) \times 20 \\ &= 850 \text{ SGD}\end{aligned}$$

**Profit with Uniform Pricing:**

$$\begin{aligned}\pi_{UP} &= (P^* - MC) \cdot Q_{total}^* \\ &= (33.33 - 10) \times 35 \\ &= 816.67 \text{ SGD}\end{aligned}$$

**Profit Difference:**

$$\Delta\pi = \pi_{PD} - \pi_{UP} = 850 - 816.67 = 33.33 \text{ SGD}$$

**Note:** The non-integer quantities (e.g.,  $Q = 18.33, 16.67$ ) are acceptable in this continuous demand model. In practice, these could represent quantities in larger units (e.g., hundreds of customers) or be rounded to whole numbers for implementation.

## Problem 2

Given Information about movie theater customers:

- Students (S):  $Q_S = 100 - 8P_S$
- Non-students (N):  $Q_N = 100 - 4P_N$
- Marginal Cost:  $MC = 2$

(a) The uniform price should be applied is showing below:

- Uniform Price:  $P^* = 9.33 \text{ SGD}$
- Total Quantity:  $Q_{total}^* = 88$  (Students: 25.36, Non-students: 62.68)

The reasoning is as follows: to find the aggregate demand, we need to sum quantities at each price level. Here set  $P_S = P_N = P$

$$\begin{aligned}Q_S &= 100 - 8P \text{ (valid for } P \leq 12.5) \\ Q_N &= 100 - 4P \text{ (valid for } P \leq 25)\end{aligned}$$

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(i) When  $P \leq 12.5$ , both groups buy:

$$Q_{total} = Q_S + Q_N = (100 - 8P) + (100 - 4P) = 200 - 12P$$

Profit function:

$$\begin{aligned}\pi(P) &= (P - MC) \cdot Q_{total}(P) = (P - 2)(200 - 12P) \\ &= 200P - 12P^2 - 400 + 24P = 224P - 12P^2 - 400\end{aligned}$$

Taking derivative with respect to P:

$$\frac{d\pi}{dP} = 224 - 24P$$

Setting  $\frac{d\pi}{dP} = 0$ :

$$224 - 24P = 0 \implies P^* = 9.33$$

Therefore:

$$\begin{aligned}Q_{total}^* &= 200 - 12 \times 9.33 = 200 - 112 = 88 \\ \pi &= (9.33 - 2) \times 88 = 7.33 \times 88 = 645.04\end{aligned}$$

The price is valid since  $P^* = 9.33 \leq 12.5$ .

(ii) When  $12.5 < P \leq 25$ , only non-students buy:

$$Q_{total} = Q_N = 100 - 4P$$

Profit function:

$$\begin{aligned}\pi(P) &= (P - MC) \cdot Q_N(P) = (P - 2)(100 - 4P) \\ &= 100P - 4P^2 - 200 + 8P = 108P - 4P^2 - 200\end{aligned}$$

Taking derivative with respect to P:

$$\frac{d\pi}{dP} = 108 - 8P$$

Setting  $\frac{d\pi}{dP} = 0$ :

$$108 - 8P = 0 \implies P^* = 13.5$$

Therefore:

$$\begin{aligned}Q_{total}^* &= 100 - 4 \times 13.5 = 100 - 54 = 46 \\ \pi &= (13.5 - 2) \times 46 = 11.5 \times 46 = 529\end{aligned}$$

The price is valid since  $13.5 \in (12.5, 25]$ .

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(iii) Boundary at  $P = 12.5$ :

At the transition point where students stop buying (the profit function has a kink due to piecewise demand):

$$Q_S = 100 - 8 \times 12.5 = 0, \quad Q_N = 100 - 4 \times 12.5 = 50$$
$$\pi = (12.5 - 2) \times 50 = 525$$

Comparing all three cases: Case (i) yields  $\pi = 645.04$ , Case (ii) yields  $\pi = 529$ , and Case (iii) yields  $\pi = 525$ . Since  $645.04 > 529 > 525$ , the optimal uniform price is  $P^* = 9.33$  from Case (i).

(b) The prices under discrimination should be applied are showing below:

- Students:  $P_S^* = 7.25$  SGD
- Non-students:  $P_N^* = 13.50$  SGD

The reasoning is as follows: for profit maximization, we apply the rule  $MR = MC$  for each market separately.

**Students:**

$$P_S = 12.5 - 0.125Q_S$$
$$R_S = P_S \cdot Q_S = (12.5 - 0.125Q_S)Q_S = 12.5Q_S - 0.125Q_S^2$$
$$MR_S = \frac{d(R_S)}{dQ_S} = 12.5 - 0.25Q_S$$

Setting  $MR_S = MC$ :

$$12.5 - 0.25Q_S = 2$$
$$0.25Q_S = 10.5$$
$$Q_S^* = 42$$

Therefore:  $P_S^* = 12.5 - 0.125(42) = 12.5 - 5.25 = 7.25$

**Non-students:**

$$P_N = 25 - 0.25Q_N$$
$$R_N = P_N \cdot Q_N = (25 - 0.25Q_N)Q_N = 25Q_N - 0.25Q_N^2$$
$$MR_N = \frac{d(R_N)}{dQ_N} = 25 - 0.5Q_N$$

Setting  $MR_N = MC$ :

$$25 - 0.5Q_N = 2$$
$$0.5Q_N = 23$$
$$Q_N^* = 46$$

Therefore:  $P_N^* = 25 - 0.25(46) = 25 - 11.5 = 13.50$

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(c) The total profits in the 2 scenarios are showing below:

- Profit with Price Discrimination: 749.50 SGD
- Profit with Uniform Pricing: 645.04 SGD
- Difference: 104.46 SGD (16.2% increase)

The calculations are as follows:

**Profit with Uniform Pricing:**

At  $P^* = 9.33$ :

$$\begin{aligned}Q_S &= 100 - 8(9.33) = 100 - 74.64 = 25.36 \\Q_N &= 100 - 4(9.33) = 100 - 37.32 = 62.68 \\Q_{total} &= 25.36 + 62.68 = 88.04 \approx 88\end{aligned}$$

Profit:

$$\begin{aligned}\pi_{UP} &= (P^* - MC) \cdot Q_{total}^* \\&= (9.33 - 2) \times 88 \\&= 645.04\end{aligned}$$

**Price Discrimination:**

Quantities:  $Q_S^* = 42$ ,  $P_S^* = 7.25$ ,  $Q_N^* = 46$ ,  $P_N^* = 13.50$

Profit:

$$\begin{aligned}\pi_{PD} &= (P_S^* - MC) \cdot Q_S^* + (P_N^* - MC) \cdot Q_N^* \\&= (7.25 - 2) \times 42 + (13.50 - 2) \times 46 \\&= 749.50\end{aligned}$$

**Profit Difference:**

$$\Delta\pi = \pi_{PD} - \pi_{UP} = 749.50 - 645.04 = 104.46 \text{ SGD}$$

**Quantity Difference:**

The total quantity sold in both scenarios is approximately the same (88 customers), but price discrimination yields a significantly higher profit due to better price targeting.

(d) Comparison and conclusions are showing below:

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| Metric               | Uniform Pricing | Price Discrimination |
|----------------------|-----------------|----------------------|
| Student Price        | SGD 9.33        | SGD 7.25             |
| Non-student Price    | SGD 9.33        | SGD 13.50            |
| Student Quantity     | 25.36           | 42                   |
| Non-student Quantity | 62.68           | 46                   |
| Total Quantity       | 88              | 88                   |
| Total Profit         | SGD 645.04      | SGD 749.50           |

### Conclusions

- (a) **Total Output:** Both strategies result in the same total quantity (88 customers). For linear demand curves with constant marginal cost, third-degree price discrimination redistributes output across markets without changing total output, as long as all markets are served under both regimes.
- (b) **Price Differences:**
  - Students pay **less** under discrimination (SGD 7.25 vs 9.33)
  - Non-students pay **more** under discrimination (SGD 13.50 vs 9.33)
- (c) **Quantity Redistribution:** Price discrimination shifts sales from non-students ( $62.68 \rightarrow 46$ ) to students ( $25.36 \rightarrow 42$ ), reflecting their different price sensitivities.
- (d) **Price Sensitivity:** Students have more elastic demand (with demand slope  $dQ_S/dP = -8$  compared to  $dQ_N/dP = -4$ ), meaning they are more price-sensitive, so they receive a lower price under discrimination.
- (e) **Profit Improvement:** Price discrimination increases profit by 104.46 SGD (16.2% increase), demonstrating its effectiveness in extracting consumer surplus from less elastic segments while expanding sales to more elastic segments.
- (f) **Welfare Implications:** Since total output remains constant, the welfare redistribution is from consumers to the firm, with no change in total surplus (deadweight loss remains the same).